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Winter wheat mapping without ground labels via automated knowledge transfer across regions and years

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Highlights

- Propose a novel method (KT-SimSeg) using a synergy of optical and SAR image time series.
- KT-SimSeg labels training data automatically with phenological similarity metrics.
- KT-SimSeg shows high accuracies across seven regions worldwide and three years.
- KT-SimSeg outperforms model transfer methods by reducing spatiotemporal variability.
- KT-SimSeg identifies winter wheat accurately for all datasets as early as heading.

Abstract

Accurate and timely information on winter wheat distribution is essential for agricultural management and food security. However, automated approaches for large-scale winter wheat mapping are often hindered by the scarcity of training data. Moreover, well-trained classification model is usually applicable to specific spatial and temporal scales. This study proposed an automated knowledge transfer approach based on adaptive segmentation of phenological similarity images (KT-SimSeg), to shed the reliance on ground labels and reduce spatiotemporal variability for cross-region/year winter wheat mapping. The optical and radar phenological patterns of winter wheat, constructed by the prior knowledge in source domains, were transferred to target domains and aligned with the temporal sequences of undefined pixels. Two-layer phenological similarity between winter wheat and undefined pixels were calculated as indicators to label undefined pixels automatically based on an adaptive threshold segmentation algorithm. Resulting labeled pixels were further refined and were used to pre-train a random forest (RF) model for winter wheat mapping. The performance of the KT-SimSeg approach was assessed in seven regions across the globe for three years, and further compared with two model transfer approaches based on RF (MT-RF) and one-class support vector machine (MT-OCSVM) classifier. The proposed approach performed well for winter wheat mapping with F1-score values of 0.925 and 0.914 across regions and years, which was superior

Additionally, the KT-SimSeg could identify winter wheat accurately as early as the heading stage. The proposed approach offers a viable solution to produce high-quality regional winter wheat products without local ground labels, and has potential for knowledge transfer across crop production regions and years in quantitative remote sensing modeling.

Keywords

Winter wheat mapping; Phenology; Knowledge transfer; Random Forest; Across regions and years

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Introduction

As one of the most important cereal crops, wheat plays a crucial role in global food security, accounting for 30% of the global grain area and 25% of its production (FAOSTAT, 2023). Accurate and timely information regarding the spatial distribution of wheat is crucial for monitoring crop growth and predicting yields (Wang et al., 2019). With the capacity to periodically and consistently observe the Earth, satellite remote sensing has been an indispensable tool for monitoring agricultural land dynamics for decades (Defourny et al., 2019). Up to now, a large number of remote sensing-based winter wheat maps have been generated with great efforts, but they generally cover a limited spatial extent or exhibit a coarse spatial resolution (Qiu et al., 2017, Xu et al., 2020). The existing algorithms for producing winter wheat maps are hard to generalize over space and time. There are urgent demands of high-quality and automated approach for producing large-area winter wheat products, thus improving the crop production and reducing food insecurity in the context of the changing Earth.

In recent years, the available remote sensing imagery has been continuously increasing, and therefore greatly promotes the development of wheat or other crop mapping algorithms (Liu et al., 2023, Wang et al., 2019). In general, the algorithms for winter wheat or crop type mapping can be categorized into two approaches: phenology-based and data-driven (Chen et al., 2023). Given the discrepant dynamics in phenological metrics during the growing cycle, phenology-based algorithms are frequently used for specific crop type mapping (Chen et al., 2023, Qiu et al., 2017). By detecting the distinguishable features from multi-source satellite imagery in the phenological profile, these approaches could separate the target from the non-target crop with a proper threshold. For example, Wardlow et al. (2007) demonstrated the feasibility of Normalized Difference Vegetation Index (NDVI) time series in separating winter wheat from other crops in the U.S. Central Great Plains with timing-specific threshold values. To highlight winter wheat signals, a spectral-temporal index, Combining variations Before and After estimated Heading dates (CBAH), was constructed using the Enhanced Vegetation Index (EVI) information during seeding, heading and harvesting stages, and then an empirical threshold value was determined by ground truth data for winter wheat mapping (Qiu et al., 2017). As the performance of these approaches is heavily subjected to the selected threshold, they are also defined as threshold-based methods (Xiong et al., 2024). The optimal threshold value for the approaches is normally empirical or region-specific. Generally, crop spectral, polarized and phenological stages vary between genotypes, environmental conditions and management practices. Therefore, the accuracy of phenology-based approaches often varies substantially across space and time, particularly over a large region (Wang et al., 2019).

The data-driven approaches, also known as machine learning (ML)- or classifier-based, have been regarded as the most popular and reliable solution for crop type mapping to date. Various high-performance supervised classifiers represented by random forest (RF) have been

sensing imagery (Ge et al., 2021, Xiong et al., 2024). While Xu et al. (2020) suggested the RF could identify winter wheat with an overall accuracy of 0.93 in Shandong, China, the convolutional neural network (CNN) demonstrated a comparable performance for winter wheat mapping at the provincial scale (Lei et al., 2021). However, such data-driven algorithms rely heavily on high-quality and within-season ground labels which are normally collected by a large number of field surveys, making crop type mapping time-consuming, resource-intensive and hard to generalize over regions and years (Chen et al., 2015, Lin et al., 2022). To surmount the limitation of lacking training data for crop type mapping, several approaches have been investigated and can be summarized into two mainstreams: (a) automated training data extraction and (b) model transfer.

Along with the increasing availability of prior knowledge, such as crop type mapping products or crowdsourced datasets, the knowledge-based approaches demonstrate great potentials for automated training data generation. For example, Yan and Ryu (2021) regarded the Google Street View images carrying detailed geographic coordinates and date information as a feasible proxy for fetching training data of various crop types. By means of the shift-invariant topology among crops in feature space, Lin et al. (2022) extracted the high-quality labels from Cropland Data Layer (CDL) and CDL-like products. Xiong et al. (2024) employed trajectory data from agricultural machinery to label winter wheat samples automatically. However, the application of knowledge-based approaches is often limited by the absence of such prior knowledge in a large portion of the world. Recently, another form of knowledge-based approaches attempts to extract high-confidence crop type pixels from phenology-based rules and pre-train supervised model for crop type mapping (Yang et al., 2023, Zhang et al., 2022). Although such idea reduces the dependence on public datasets, the rules for separating crop types are hard to generalize across regions and years considering the substantial inter- and intra-class spectral and phenological variability.

Pre-training a classification model with abundant prior knowledge in source region/year, cross-domain model transfer (MT) become a popular solution for producing high-quality crop type mapping products in the sample-deficient region/year. You and Dong (2020) established RF models with sufficient field samples in the previous year and accurately identified the rice, soybean and corn in the next year by cross-year MT. Extracting sufficient corn and rice samples from CDL, Ge et al. (2021) constructed a U-Net model in USA and confirmed the spatial generalization performance of MT in China. No matter using ML or DL-based MT, its performance is subjected largely to the similarity of growing conditions (Wang et al., 2019). In view of the signature shift between source and target domain, most transfer learning techniques (e.g., fine-turning-based, multi-task or few-shot) required training labels in target domain for model fine-turning (Ma et al., 2024). However, as mentioned above, ground labels are often difficult to be obtained, particularly in most developing countries.

For producing winter wheat products automatically and seamlessly in space and time, knowledge transfer is promising by addressing the absence of training data, but cross-domain spatiotemporal variability should to be mitigated. Several studies have revealed that the time-weighted dynamic time warping (TWDTW) algorithm has weak dependence on training data and tolerance in phenological distorts for crop mapping (Belgiu and Csillik, 2018, Gella et al., 2021). With the standard temporal curve of NDVI from each province, Dong et al. (2020a) extracted the winter wheat successfully in China for three consecutive years. The effectiveness and efficiency of phenology-TWDTW (PTDTW) were also revealed in Dong et al. (2020b) for regional winter wheat mapping. Similar to most phenology-based approaches, these algorithms also require empirical and region-specific similarity thresholds for specific crop type mapping, which hinders the application of the approach over a large scale. Additionally, the feasibility of TWDTW on cross-domain winter wheat mapping with prior knowledge in different agro-ecological regions around the world has not been evaluated.

mapping around the world using local prior knowledge directly; (2) to develop an automated knowledge transfer approach for labeling training samples with optical and radar indicators retrieved from TWDTW; (3) to evaluate the effectiveness and scalability of the approach across regions and years for winter wheat mapping over various agro-ecological regions around the world.

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Study area

Seven test regions (TRs) with each covering an area of 10,000 km^2 distributed in five major winter wheat-producing countries across Asia, Europe and North America (Fig. 1 **& Fig. S1**) were selected to test the effectiveness and scalability of the proposed approach. With diverse climate conditions, crop types, field sizes, crop calendars and cropping systems (Table 1), these regions are representative to test the effectiveness and scalability of the proposed framework.

The first and second test …

Winter wheat mapping with local prior knowledge

Based on the seed samples, Fig. 5 exhibits the performance of winter wheat mapping with various features and approaches. Generally, combing all optical or SAR features achieved higher accuracy than single feature, and stacking all S1 and S2 features obtained the highest performance no matter whether using TWDTW, RF or OCSVM. Among the three approaches over four TRs, the RF achieved the best performance for winter wheat mapping (OA & F1=0.980 & 0.984), followed by OCSVM (OA & F1=0.959 & …

Advantages of the KT-SimSeg approach

At present, the major constraint for large-area crop type mapping is ascribed to the shortage of ground labels (Wang et al., 2019, Yan and Ryu, 2021). Recently, a number of approaches were proposed to extract training samples automatically by means of local prior knowledge, such as region-specific crop type mapping products (Lin et al., 2022), trajectory data from agricultural machinery (Xiong et al., 2024), crop rotation (Zhang et al., 2021) or specific phenological information (Yang et al., …

Conclusions

A reliable and robust approach for automated crop type mapping would be of value to the crop production and food security. Supported by time-series Sentinel-1 and Sentinel-2 images, this research designed an automated knowledge transfer approach for winter wheat mapping

CRediT authorship contribution statement

Gaoxiang Yang: Writing – original draft, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Xingrong Li:** Investigation, Conceptualization, Methodology. **Yuan Xiong:** Investigation, Conceptualization. **Meng He:** Investigation. **Xue Wang:** Funding acquisition. **Xia Yao:** Conceptualization. **Yan Zhu:** Funding acquisition, Conceptualization. **Weixing Cao:** Funding acquisition, Formal analysis, Conceptualization. **Tao Cheng:** Writing – review & editing, Writing – original draft, …

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. …

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